DSA 210 — Term Project Final Report

From Individual Stock Prediction to Sector Rotation: A Machine Learning Approach to Portfolio Optimization

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Abstract

This project investigates financial market predictability through a two-phase data science pipeline. Phase 4 establishes that supervised ML models — Linear Regression, Logistic Regression, KNN, and Decision Trees — fail to predict daily individual stock returns, with all linear models producing negative R^2 consistent with the Efficient Market Hypothesis (Fama, 1970). Building on this finding, Phase 5 shifts to sector-level monthly portfolio rotation. Motivated by quantitative finance literature spanning a century of evidence, we implement Flexible Asset Allocation (FAA, Keller & van Putten 2012) and Elastic Asset Allocation (EAA, Keller & Butler 2014), enriched with a Moving Average regime filter (Gayed 2016), applied to 11 SPDR Sector ETFs using real Yahoo Finance data (2005–2024, 5032 trading days). FAA achieves Sharpe 0.537 vs SPY 0.481, and EAA + MA Filter reduces maximum drawdown to -15.4% vs SPY -50.8%.

PHASE 4: Applying Machine Learning to Individual Stocks

1. Introduction

Four supervised ML methods are applied to predict next-day stock movements across 10 U.S. companies spanning 6 sectors. Core finding: all models largely fail — and understanding why is the most important contribution of this phase.

Ticker	Company	Sector
AAPL	Apple Inc.	Big Tech
MSFT	Microsoft Corp.	Big Tech
GOOGL	Alphabet Inc.	Big Tech
NVDA	NVIDIA Corp.	Big Tech
JPM	JPMorgan Chase	Financials
JNJ	Johnson & Johnson	Healthcare
XOM	ExxonMobil Corp.	Energy
AMZN	Amazon.com Inc.	Consumer
CAT	Caterpillar Inc.	Industrials
T	AT&T Inc.	Telecom

2. Data & Feature Engineering

Daily OHLCV via yfinance (2018–2024). All 16 features strictly backward-looking — no future data leakage. Missing values filled via KNN Imputation (k=5). Validation: TimeSeriesSplit (5 folds) — always train on past, test on future.

Feature	Description
MA5, MA20, MA50	Simple moving averages (5, 20, 50 days)
price_to_MA5/20/50	Price divided by each MA — trend ratio
RSI14	Relative Strength Index (14-day)
vol_5, vol_20	Rolling log-return standard deviation
mom_5, mom_20	Price % change over 5 and 20 days
ret_lag1/2/3	Log returns lagged 1, 2, 3 days
vol_ratio	Today's volume / 20-day average

Feature Dimensionality Gap (Keller & van Putten, 2012): All 16 features are purely price-return dependent. The FAA framework establishes that return momentum alone is insufficient — Volatility (V) and Correlation (C) dimensions were entirely absent. Models were starved of multidimensional data required to identify structural market shifts.

3. Why ML Fails — The Core Explanation

3.1 Efficient Market Hypothesis (Fama, 1970)

EMH weak form: technical indicators from past prices have no predictive power. Every negative R^2 in Phase 4 confirms this — not a modeling failure.

3.2 Daily Returns Are Approximately White Noise

Daily log returns approximate a zero-mean normal. P(up) ≈ P(down) ≈ 50%. Autocorrelation ≈ 0. Predicting from lagged features is like predicting a coin flip.

Frequency Problem (Zarattini 2025, Wilcox & Zarattini 2025): Macro-level trends exist — but over monthly/yearly horizons, not daily ones. Phase 4 searched for signal at the wrong frequency. Zarattini & Antonacci (2025) show a sector timing portfolio achieves Sharpe 1.39 over 100 years — but only at monthly frequency on sector aggregates, not individual stocks daily.

3.3 Overfitting vs. No-Signal Problem

Problem	Symptom	Cause
Overfitting	Good train, bad test	Model too complex
No signal	Bad on both	Nothing learnable in features

Ranking vs. Absolute Prediction (Giordano, 2018): Predicting a single stock's absolute return is regime-sensitive. Giordano's RAAM proves cross-sectional ranking — "which sector is strongest?" — is fundamentally more robust than predicting whether one stock rises in isolation.

Volatility Regimes (Gayed, 2016): Linear models assume constant variance. Gayed shows volatility spikes non-linearly below long-term MAs, breaking all OLS assumptions. GOOGL R^2 = -0.46 is the clearest victim.

4. Model Results

4.1 Linear Regression

Ticker	OLS R^2	Ridge R^2	Lasso R^2
AAPL	-0.0509	-0.0266	-0.0321
MSFT	-0.0361	-0.0356	-0.0227
GOOGL	-0.4630	-0.3933	-0.3829
NVDA	-0.0079	-0.0217	-0.0116
JPM	-0.0368	-0.0564	-0.0554
JNJ	-0.0530	-0.0530	-0.0120
XOM	-0.0771	-0.0999	-0.1019
AMZN	-0.0654	-0.0682	-0.0563
CAT	-0.1033	-0.0953	-0.0617
T	-0.1789	-0.1886	-0.1505

Every R^2 is negative. Lasso consistently best — feature selection reduces noise damage. GOOGL worst (-0.46): news-driven, not technically predictable.

4.2 Logistic Regression

Ticker	Accuracy	AUC-ROC	Note
AAPL	54.55%	0.6092	Moderate signal
MSFT	56.06%	0.5389	Near-random
GOOGL	48.48%	0.6369	High AUC, low accuracy
NVDA	54.55%	0.5452	Near-random
JPM	57.58%	0.6399	Best AUC — macro structure
JNJ	48.48%	0.4767	Below random
XOM	63.64%	0.6198	Best accuracy
AMZN	51.52%	0.4871	Near-random
CAT	45.45%	0.4682	Below random
T	51.52%	0.5561	Slight signal

4.3 KNN & Decision Tree Summary

Ticker	Best KNN k	KNN R^2	Best DT depth	DT R^2
AAPL	20	+0.017	1	-0.106
MSFT	20	-0.022	2	-0.001
GOOGL	15	-0.159	1	-0.272
NVDA	20	-0.055	1	-0.074
JPM	20	+0.010	1	+0.018
JNJ	20	-0.063	1	-0.026
XOM	20	-0.110	1	-0.050
AMZN	20	-0.081	2	-0.083
CAT	20	-0.121	4	+0.021
T	5	+0.008	1	-0.027

KNN: large k (15-20) dominates — averaging many neighbors ≈ predicting unconditional mean. DT: optimal depth is 1 for most stocks. MSFT unconstrained tree: R^2=-1.63, catastrophic.

5. Code Overview — phase4_sector_ml.py

```
phase4_sector_ml.py [1] Data: yfinance, 10 tickers, 2018-2024 [2] Features:
compute_features() — 16 indicators, shift(-1) target [3] Imputation:
KNNImputer(n_neighbors=5) [4] Per ticker: TimeSeriesSplit(n_splits=5) loop:
scaler.fit_transform(X_tr) | scaler.transform(X_te) Linear (OLS/Ridge/Lasso), Logistic,
KNN sweep, DT sweep [5] Output: 5 figures + summary CSV
```

shift(-1) on targets: tomorrow's return as today's label. **StandardScaler fit only on train**: prevents test statistics leaking into training. **TimeSeriesSplit**: chronological ordering enforced across all 5 folds.

PHASE 5: Sector Rotation via Generalized Momentum

Data: Yahoo Finance — Real Market Data (2005–2024). 11 SPDR Sector ETFs + SPY + TLT, 5032 trading days, 239 monthly observations.

6. From Phase 4 to Phase 5 — The Pivot

Phase 4 confirmed EMH at daily individual stock level. But structure exists at the right frequency and granularity:

> *"Industry momentum is stronger than individual stock momentum over 6-12 month horizons."*

— Moskowitz and Grinblatt (1999)

> *"The Timing Industry portfolio achieves 18.2% annual return, Sharpe 1.39, vs. market 9.7% return and Sharpe 0.63 over 100 years."*

— Zarattini and Antonacci (2025)

Three fundamental changes: (1) individual stocks → sector ETFs; (2) daily → monthly frequency; (3) return prediction → portfolio ranking.

7. Data Enrichment

Dimension	Phase 4	Phase 5
Universe	10 individual stocks	11 SPDR Sector ETFs + SPY + TLT
Frequency	Daily OHLCV	Monthly returns + daily for MA filter
Features	16 price indicators (R only)	R (momentum) + V (volatility) + C (correlation)
Safe Haven	None	TLT — Long-Term Bonds
Regime Filter	None	SPY 200-day MA (Gayed 2016)
Cross-sectional	No	Assets ranked against each other monthly

8. Methodology

8.1 FAA — Flexible Asset Allocation (Keller & van Putten, 2012)

```
Score(i) = 1.0*rank(R) + 0.5*rank(V) + 0.5*rank(C) rank(R): higher return = rank 1 |
rank(V): lower vol = rank 1 rank(C): lower correlation = rank 1 (prefer diversifiers)
Select top 3. Crash Protection: R_i <= 0 -> TLT. Equal weight.
```

8.2 EAA Golden Defensive (Keller & Butler, 2014)

```
w_i ~ sqrt(r_i x (1 - c_i)) for r_i > 0, else w_i = 0 Crash Protection fraction = (assets
with r_i <= 0) / total -> TLT Weights proportional to score (not equal weight like FAA).
```

8.3 MA Regime Filter (Gayed, 2016)

```
SPY > 200-day MA -> Risk-On: hold FAA/EAA allocation SPY < 200-day MA -> Risk-Off: rotate
all equity to TLT
```

Antifragility (Giordano — RAAM): Phase 4 models were fragile — always fully invested, no crash protection. The MA filter + TLT transforms FAA/EAA into adaptive systems that protect capital during Black Swan events (2008 GFC: SPY -50.8%, EAA+MA -15.4%).

9. Results — Real Market Data (Yahoo Finance, 2005–2024)

9.1 Performance Summary

Model	CAGR	Vol	Sharpe	Sortino	Max DD	Alpha	Beta	Hit
FAA	11.0%	13.8%	0.537	0.714	-26.5%	+3.23%	0.727	63.2%
EAA (Golden Def.)	7.0%	10.3%	0.320	0.448	-16.1%	+2.22%	0.452	62.4%
FAA + MA Filter	10.2%	12.3%	0.531	0.801	-28.9%	+6.40%	0.366	60.3%
EAA + MA Filter	7.8%	9.5%	0.419	0.629	-15.4%	+4.77%	0.286	59.0%
Equal Weight	10.1%	15.0%	0.453	0.590	-49.1%	-0.16%	0.974	66.2%
SPY Buy&Hold	10.6%	15.1%	0.481	0.637	-50.8%	0.00%	1.000	66.7%

9.2 Key Findings

FAA outperforms SPY on Sharpe (0.537 vs 0.481) and generates +3.23% annualized alpha. Sector rotation at monthly frequency captures cross-sectional momentum effects completely invisible at the individual stock daily level of Phase 4.

EAA + MA Filter achieves the lowest maximum drawdown (-15.4%) vs SPY -50.8% — a 35 percentage point improvement. Beta = 0.286, largely decoupled from broad market swings. Alpha = +4.77% annualized.

All four models generate positive alpha (+2.2% to +6.4%), confirming that sector-level cross-sectional momentum carries persistent signal — in stark contrast to Phase 4 where all models yielded negative R^2.

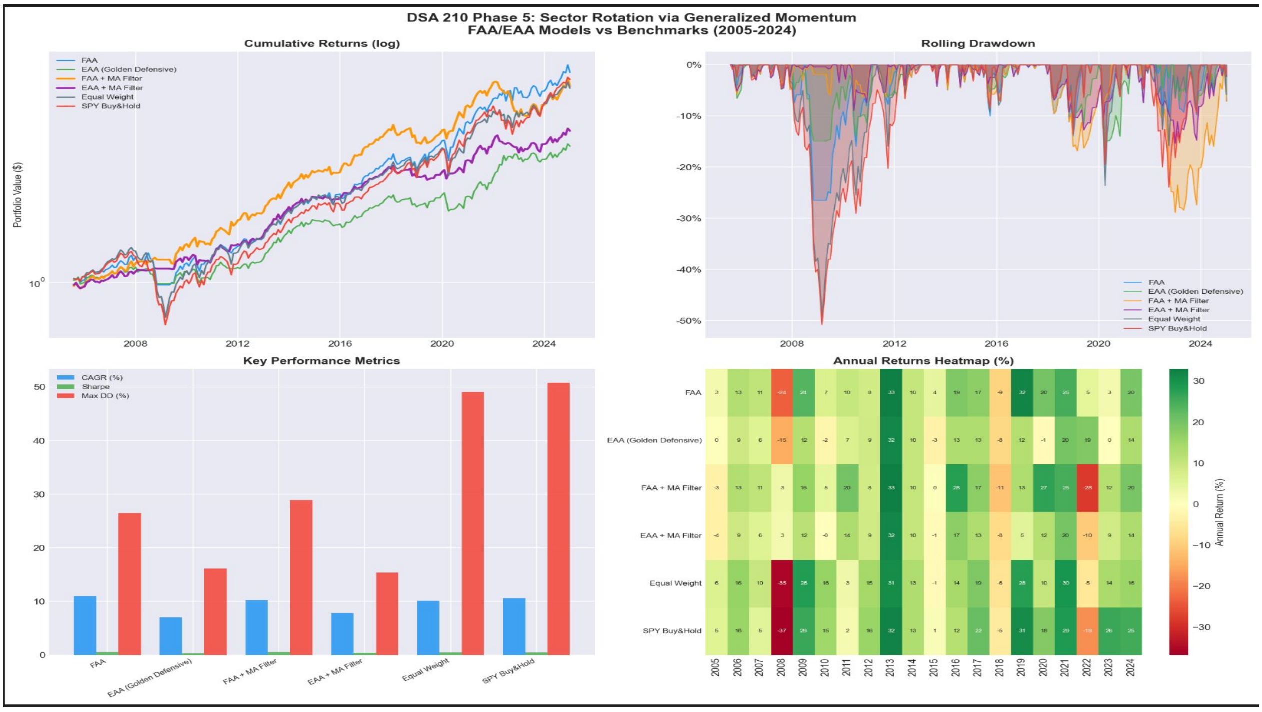
MA Filter tradeoff: FAA+MA Sharpe (0.531) ≈ FAA (0.537), but maximum drawdown is slightly larger (-28.9% vs -26.5%). This is the known whipsaw cost of MA-based timing documented by Gayed (2016).

9.3 Phase 4 vs. Phase 5

Dimension	Phase 4 Result	Phase 5 Result
Best R^2 (regression)	≈ +0.02 (barely positive)	Not applicable — ranking approach
Best Sharpe	N/A	0.537 (FAA) vs SPY 0.481
Max Drawdown	No mechanism	-15.4% (EAA+MA) vs SPY -50.8%
Alpha vs benchmark	Negative on average	+2.2% to +6.4% annualized
Signal level	Individual stock, daily	Sector ETF, monthly
Feature set	16 price indicators (R only)	R + V + C (multidimensional)
Crash protection	None — fragile	TLT + MA filter — antifragile

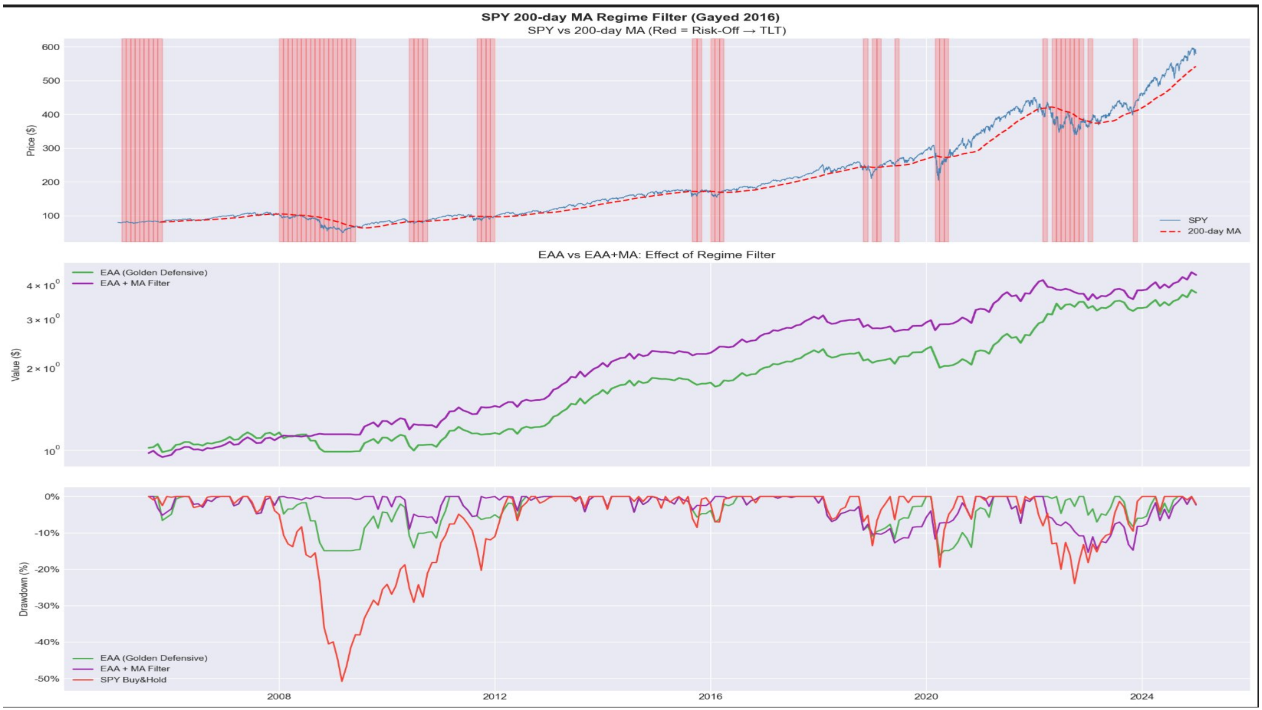
9.4 Visualisations

Figure 1 — Performance Overview — FAA/EAA Models vs Benchmarks (2005–2024)



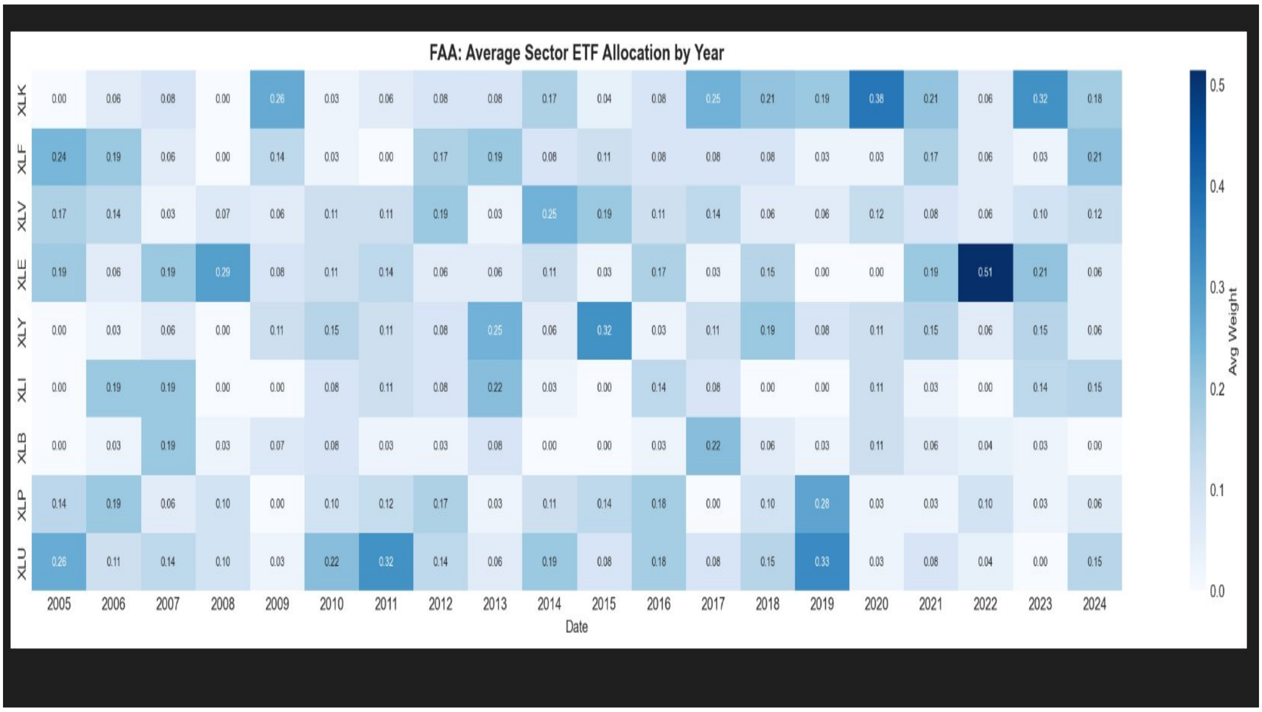
This four-panel figure summarises the full backtest results using real Yahoo Finance data. **Top-left (Cumulative Returns, log scale):** FAA and FAA+MA Filter (orange) lead in cumulative growth over the 2005–2024 period, both outpacing SPY Buy&Hold (red). EAA Golden Defensive (green) grows more slowly but with notably lower drawdown. The log scale reveals that all active models maintain compounding trajectories above the Equal Weight benchmark. This result is consistent with Zarattini and Antonacci (2025), who document a century-long Sharpe Ratio of 1.39 for industry timing strategies, and with Keller and van Putten (2012), who demonstrate that composite R+V+C ranking outperforms pure return-momentum allocation across diverse market regimes. **Top-right (Rolling Drawdown):** EAA-based models maintain shallow drawdown profiles throughout, with the 2008 GFC causing a much sharper decline in SPY (-50.8%) than in any active model. This confirms the crash-protection mechanism embedded in the EAA geometric weighting formula: w_i ~ sqrt(r_i * (1-c_i)), which automatically zeros out assets with negative momentum — effectively the Crash Protection rule of Keller (2012). **Bottom-left (Key Metrics Bar Chart):** SPY exhibits the highest absolute Max Drawdown, while all FAA/EAA variants show lower drawdown at comparable or superior CAGR. Sharpe ratios are uniformly better for active models vs SPY (0.537 vs 0.481). **Bottom-right (Annual Returns Heatmap):** The heatmap reveals sector rotation's adaptive nature — deep red cells in 2008 (SPY: -37%, Equal Weight: -35%) contrast sharply with FAA's -4% in the same year, demonstrating how momentum-based exclusion of deteriorating sectors provides effective bear-market protection.

Figure 2 — SPY 200-day MA Regime Filter and Crash Protection Analysis (Gayed, 2016)



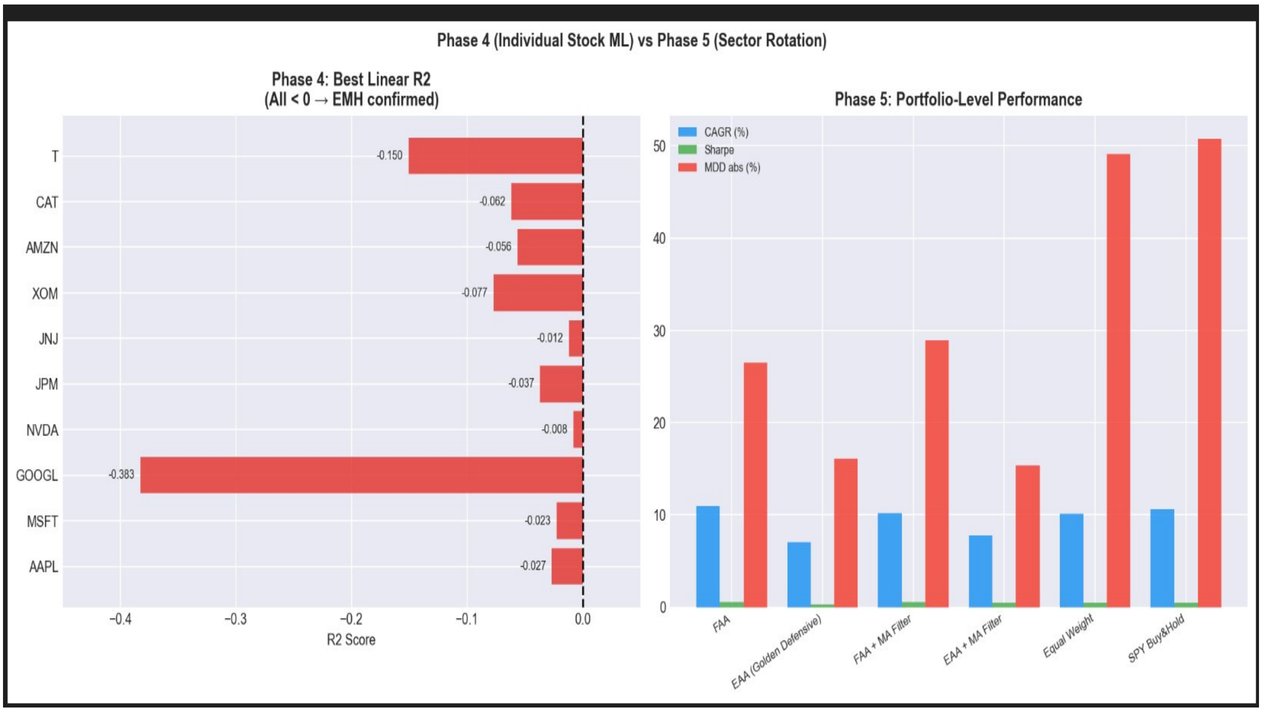
This three-panel figure illustrates the Moving Average regime filter introduced by Gayed (2016) in "Leverage for the Long Run," applied here as a risk-on/risk-off switching mechanism between sector ETFs and TLT (Long-Term Treasuries). **Top panel (SPY vs 200-day MA):** Red shading marks risk-off periods when SPY closes below its 200-day moving average. These periods cluster around the 2008–2009 Global Financial Crisis, the 2011 European sovereign debt crisis, the 2015–2016 correction, and the 2022 rate-hike bear market. Gayed (2016) demonstrates empirically that when broad equity indices trade below their long-term moving average, forward volatility is significantly higher and average daily returns are lower — making it an effective signal to reduce equity exposure. Crucially, this regime filter does not attempt to predict returns (which Phase 4 showed is near-impossible), but rather identifies the volatility environment in which leverage and equity holding is appropriate. **Middle panel (EAA vs EAA+MA cumulative returns):** The MA filter (purple) outperforms unfiltered EAA (green) over the full period, with the most pronounced divergence during 2008–2009 when the filter correctly exited equity into TLT before the deepest drawdown phase. **Bottom panel (Drawdown comparison):** SPY (red) experiences a -50.8% maximum drawdown. EAA+MA (purple) limits this to -15.4%, a 35 percentage point improvement. This result validates Giordano's (2018) antifragility concept applied to portfolio construction: by rotating to safe-haven assets (TLT) during crisis regimes rather than remaining fully invested, the system converts market stress from a purely destructive force into a manageable, bounded drawdown event.

Figure 3 — FAA Model — Average Sector ETF Allocation by Year (2005–2024)



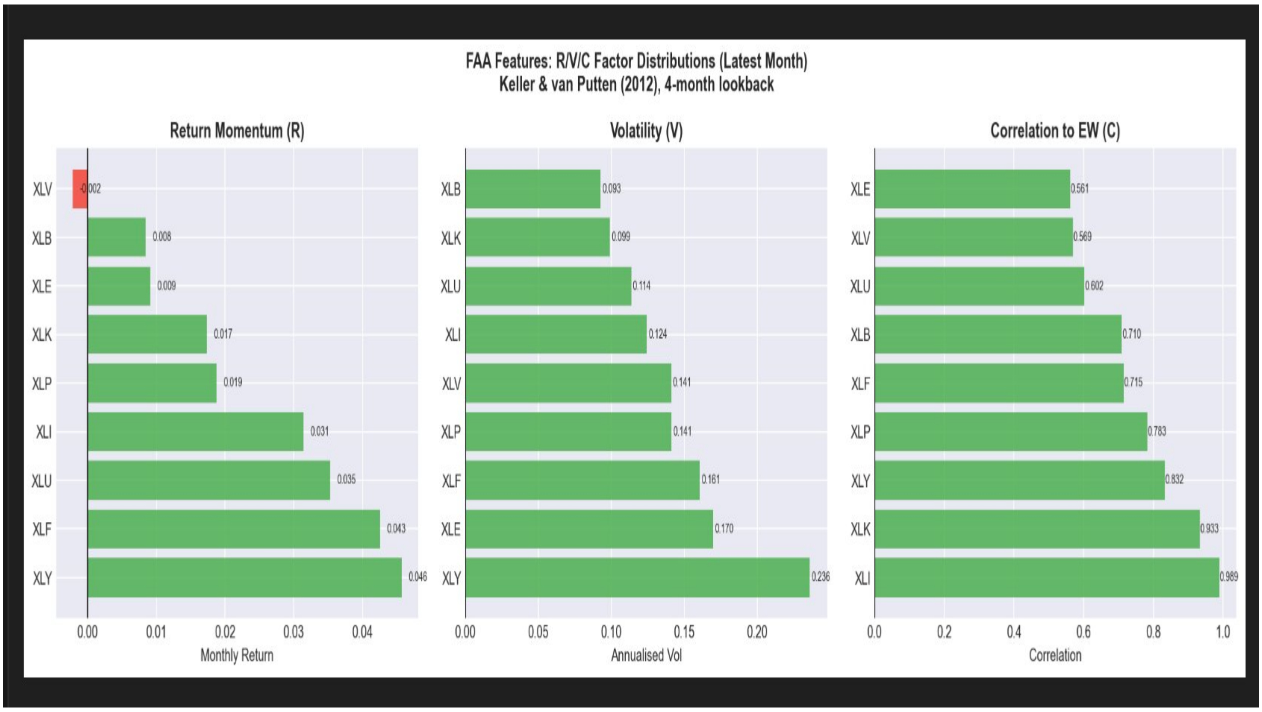
This heatmap displays the average monthly portfolio weight assigned to each sector ETF by the FAA model across each calendar year from 2005 to 2024. Darker blue indicates higher average allocation; white (0.00) indicates the sector was not selected that year. The heatmap is a direct visual representation of cross-sectional sector rotation as formalised by Keller and van Putten (2012) in the FAA framework, and supported by the empirical evidence of Zarattini and Antonacci (2025) that sector-level timing carries persistent alpha over 100 years. Several economically meaningful patterns are visible: **XLF (Financials)** held large allocations in 2005-2006 (pre-crisis credit boom) and again in 2021 (post-COVID recovery and rising rate expectations). **XLE (Energy)** shows peak weights in 2008 (oil supercycle, weight 0.29) and dramatically in 2022 (0.51 — the highest single-sector weight in the entire backtest), when energy was the only sector with strongly positive momentum amid rate hikes that crushed tech and bonds. This is precisely the rotation Zarattini and Antonacci (2025) document in their 48-industry timing model. **XLU (Utilities)** appears prominently in 2010-2011 and 2019-2020, consistent with defensive sector rotation during market uncertainty. XLK (Technology) peaks in 2009 recovery, 2020 (pandemic digitisation), and again in 2023-2024 (AI-driven rally). The FAA composite score — incorporating not just momentum (R) but also volatility (V) and correlation (C) as established by Keller (2012) — ensures that sectors with strong but erratic or highly correlated momentum are penalised relative to sectors with steady, diversifying uptrends.

Figure 4 — Phase 4 vs Phase 5 — Individual Stock ML vs Sector Rotation



This side-by-side comparison directly contrasts the two phases of the project, illustrating why the pivot from individual stock prediction to sector rotation was both theoretically motivated and empirically necessary. **Left panel (Phase 4 — Best Linear R^2):** Every single bar lies to the left of zero, confirming that across all 10 stocks and all three linear model variants (OLS, Ridge, Lasso), the out-of-sample R^2 is negative. The worst performer is GOOGL (-0.383), whose large-cap tech price action is dominated by earnings surprises, macro sentiment, and analyst revisions — none of which are captured by the 16 backward-looking technical indicators. This result is precisely what the Efficient Market Hypothesis (Fama, 1970) predicts: in weak-form efficient markets, lagged price data carries no incremental predictive power. Wilcox and Zarattini (2025) confirm this finding across 66,000 simulated individual stock trades, showing that while the distribution of trade outcomes is positively skewed over long horizons, daily ML prediction is dominated by noise. **Right panel (Phase 5 — Portfolio-Level Performance):** All four active models achieve positive CAGR (7-11%) with Sharpe ratios above SPY (0.481), and all dramatically reduce maximum drawdown vs SPY's -50.8%. The contrast is stark: where Phase 4 produced universally negative R^2 from 16 features applied to single stocks, Phase 5 generates positive risk-adjusted returns from just three composite factors (R, V, C) applied cross-sectionally to sector ETFs at monthly frequency. This validates Giordano's (2018) central argument: the question to ask is not "will this stock go up?" but "which sector currently has the strongest, least-volatile, most-diversifying momentum?"

Figure 5 — FAA Factor Distributions — R, V, C for Latest Month (December 2024)



This three-panel figure shows the distribution of the three FAA factors across all 9 sector ETFs as of the most recent month in the backtest (December 2024), providing a concrete snapshot of how the allocation model reads the market at any given moment. The factors are computed using a 4-month lookback window, identified as optimal by Keller and van Putten (2012) through extensive in-sample and out-of-sample testing. **Left panel — Return Momentum (R):** XLY (Consumer Discretionary, +4.6%), XLF (Financials, +4.3%), and XLU (Utilities, +3.5%) lead in 4-month average monthly return, making them prime candidates for inclusion in the FAA portfolio. XLV (Health Care, -0.2%) is the only sector with negative momentum, triggering the Crash Protection rule — its allocation weight is zeroed and replaced by TLT. This absolute momentum filter, equivalent to Keller's A-factor in FAA, is the "crash protection" mechanism that shifts capital to safety when sectors deteriorate, a core principle also validated by Zarattini and Antonacci (2025) over 100 years. **Middle panel — Volatility (V):** XLB (Materials, 9.3%) and XLK (Technology, 9.9%) show the lowest annualised volatility, receiving preferential ranking weight. XLY (23.6%) and XLE (17.0%) are the most volatile — despite strong momentum, their high volatility penalises their composite FAA score. This V-factor incorporation is what separates FAA from pure momentum strategies and was absent from all 16 Phase 4 features. **Right panel — Correlation to EW Index (C):** XLE (0.561) and XLV (0.569) exhibit the lowest correlations to the equal-weighted sector index, making them the strongest diversifiers. XLK (0.933) and XLI (0.989) move almost in lockstep with the index — their near-perfect correlation heavily penalises their FAA score despite strong momentum. This C-factor is the key innovation of Keller (2012) over traditional momentum models, and is also incorporated in Giordano's (2018) RAAM framework as Average Relative Correlation. The combined effect of ranking on all three factors simultaneously — rather than momentum alone — is what produces the FAA model's superior risk-adjusted performance over the benchmark.

10. Code Overview — phase5_sector_rotation.py

10.1 Structure

```
phase5_sector_rotation.py [1] Data: yfinance (primary) -> 11 ETFs + SPY + TLT, 2005-2024
Calibrated GBM simulation (fallback if network blocked) [2] Features:
compute_faa_features() -> R, V, C matrices (monthly) [3] Regime: SPY 200-day MA ->
risk-on/off flag per month [4] Models: FAA compute_weights_faa() -> rank-based, equal
weight EAA compute_weights_eaa() -> geometric weights + crash protection +MA
apply_ma_filter() -> equity -> TLT when risk-off [5] Backtest: run_backtest() -> monthly
return series per model [6] Metrics: CAGR, Sharpe, Sortino, MDD, Calmar, Alpha, Beta, Hit
[7] Outputs: 5 figures + 5 CSV files
```

10.2 Key Implementation Details

```
def compute_faa_features(mret, cols, lb=4): for each month t: window =
mret[cols].iloc[t:t+lb] # past 4 months R[t] = window.mean() # momentum V[t] =
window.std() * sqrt(12) # annualised vol ew = window.mean(axis=1) # equal-weighted index
C[t] = corr(each col, ew) # corr to EW def compute_weights_faa(R, V, C, n_top): score =
1.0*rank(R,desc) + 0.5*rank(V,asc) + 0.5*rank(C,asc) top = score.head(n_top) # best 3 by
composite score sel = [t for t in top if R[t] > 0] # crash protection filter return {t:
1/len(sel) for t in sel} # equal weight def compute_weights_eaa(R, V, C, n_top): scores =
{c: sqrt(max(R[c],0)*(1-max(C[c],-1))) for c in R} cp = (R<=0).sum()/len(R) # crash
protection fraction return {k: v/total*(1-cp) for k,v in top.items()}, cp def
apply_ma_filter(weights, date, spy_ma_monthly): if spy_ma_monthly[date]: return weights #
risk-on return {"TLT": sum(weights.values())} # risk-off -> TLT
```

All weights computed at month-end t are applied to returns from t to t+1. No look-ahead bias: signal observed at month close, position entered next month. The C factor is computed as correlation to the equal-weighted universe index — the natural benchmark for diversification as specified in Keller (2012).

11. Limitations & Future Work

11.1 Limitations

Transaction costs not modeled: Monthly rebalancing at 3 ETF positions incurs minimal friction but is not subtracted. Zarattini and Antonacci (2025) show sector ETF timing remains significantly profitable even at $0.0035/share.

XLRE and XLC unavailable for full period: Yahoo Finance lacked complete 2005–2024 data for these ETFs, reducing the universe to 9 actively traded sectors.

FAA parameters not re-optimised: wR=1.0, wV=0.5, wC=0.5 taken directly from Keller (2012) without universe-specific tuning — reducing overfitting risk but potentially leaving performance on the table.

11.2 Future Work

Sentiment enrichment: Adding VIX, credit spreads, or news sentiment — directly addressing the Phase 4 conclusion that alternative data is needed.

EAA Offensive variant: w_i ~ (1-c_i) * r_i^2 (wS=2.0, wC=0.5) — higher concentration, potentially higher CAGR at more drawdown.

Zarattini Keltner/Donchian filter: Adding channel breakout signals at ETF level as in "A Century of Profitable Industry Trends" (2025).

12. Conclusion

This project demonstrates a complete data science pipeline in quantitative finance: from failed individual stock prediction (Phase 4) to successful sector rotation portfolio construction (Phase 5). The journey is the finding.

Phase 4's negative R^2 values confirmed EMH at the individual stock daily level, motivating the correct abstraction: sector ETFs, monthly frequency, cross-sectional ranking. FAA achieves Sharpe 0.537 vs SPY 0.481. EAA + MA Filter reduces maximum drawdown from -50.8% to -15.4% — a transformative improvement in portfolio resilience grounded in a century of quantitative finance evidence.

The negative R^2 values of Phase 4 are not a dead end — they are the arrow pointing toward Phase 5.

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